Sheet (4)

(1) Write a C++ program that reads an unspecified number of integers, determines how many positive and negative values have been read, and computes the total and average of the input values (not counting zeros). Your program ends with the input 0. Display the average as a floating-point number.

(2) Write a C++ program that displays the following table (note that $1\ inch = 25.4\ millimeters$):

Millimeters	Inches
2	0.078
4	0.156
...	
96	3.744
98	3.822

(3) Write a program that displays the following table (note that $1\ meter = 3.280\ feet$):

Meters	Feet
1	3.280
2	6.560
...	
14	45.920
15	49.200

(4) Write a program that displays the following tables side by side (note that $1\ inch = 25.4\ millimeters$):

Millimeters	Inches	Inches	Millimeters
2	0.078	1	25.641
4	0.156	2	51.282
...			
98	3.822	49	1256.410
100	3.900	50	1282.051

(5) Write a program that displays the following tables side by side (note that $1\ meter = 3.280\ feet$):

Meters	Feet	Feet	Meters
1	3.280	3	0.915
2	6.560	6	1.829
...			
14	45.920	42	12.805
15	49.200	45	13.720

(6) Write a C++ program to print the following table to display the tan and cot values of degrees from 0 to 60 with increments of 10 degrees. Round the value to keep four digits after the decimal point.

```
Degree      Sin/Cos      Cos/Sin
10          0.0000       inf
20          0.1736       5.6713
...
50          1.1918       0.8391
60          1.7320       0.5774
```

(7) Write a C++ program to print the following table using the exp function:

```
Number      Exponent
0           1
1           2.71828
...
9           8103.08
10          22026.5
```

(8) Write a C++ program that displays all the numbers from 1 to 500, 5 per line, that are divisible by 3 and 4. Numbers are separated by exactly one space.

(9) Write a C++ program that displays all the numbers from 300 to 400, 5 per line, that are divisible by 3 but not divisible by 6. Numbers are separated by exactly one space.

(10) Write a C++ program that receives an ASCII code (an integer between 0 and 127) and displays its character.

A sample run gives the following output;

```
Enter an ASCII code (0-127): 69
The character for ASCII code 69 is: E
```

(11) Write a program that receives a character and displays its ASCII code.

A sample run gives the following output;

```
Enter a character: D
ASCII Value of D is 68
```

(12) Write a C++ program that prompts the user to enter an uppercase letter and converts it to a lowercase letter.

(13) Design a C++ program that prompts the user to input a single character, which can be either:

- A letter (uppercase or lowercase), or
- A single-digit number (0-9).

The program should then read the input and display one of the following messages to the user, based on the character type:

- "Uppercase Letter" if the character is an uppercase letter (A-Z).
- "Lowercase Letter" if the character is a lowercase letter (a-z).
- "Number" if the character is a single-digit number (0-9).
- "Invalid Input" if the input is not a letter or a single-digit number.

(14) Write a C++ program that prints all the uppercase characters of the ASCII character table. Display 5 characters per line. Note that ASCII for $A = 65$ and $Z = 90$.

(15) Write a C++ program that reads an integer and displays its first five multiples. That is in the input number is N the program displays $\times 1$, $N \times 2$, $N \times 3$, $N \times 4$, $N \times 5$

(16) Write a C++ program that prompts the user to enter an integer from 1 to 15 and displays a pyramid like that one shown in the below figure.

```
                1
              2 1 2
            3 2 1 2 3
          4 3 2 1 2 3 4
        5 4 3 2 1 2 3 4 5
      6 5 4 3 2 1 2 3 4 5 6
    7 6 5 4 3 2 1 2 3 4 5 6 7
```